

Supplementary Material

Afforestation as a Carbon Dioxide Removal Technology in PyPSA-Eur:
Methodology for Spatial and Temporal Sequestration Rates

Alberto Alamia [co-authors]

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0.1 Afforestation as a Carbon Dioxide Removal Technology

Afforestation was first introduced in PyPSA-eur in [Fernandes et al. \(2026\)](#) as a spatially resolved carbon dioxide removal (CDR) technology. The implementation requires two inputs per network node: (i) the available land area from CORINE land cover, (ii) a CO₂ sequestration rate in tCO₂/(haa). Here is introduced a more accurate approach to the calculation of the CO₂ sequestration rates, which ensure consistency with EU LULUCF legislation, and provides a sub-annual temporal profile distributing the uptake across calendar months. This section describes the derivation of both CO₂ sequestration rates and temporal profiles from publicly available European forest inventory and remote-sensing data.

0.1.1 Annual Sequestration Rate: Pilli et al. Yield Table Library

Data source. The sequestration rates are derived from the JRC forest growth library compiled by [Pilli et al. \(2024a\)](#), which provides age-class-resolved volume and increment curves for 222 forest types across 25 EU Member States (EU-27 excluding Cyprus and Malta). The library is distributed via Zenodo [Pilli et al. \(2024b\)](#) described in [Pilli et al. \(2024a\)](#) and contains three key datasets:

- **Standing stock** ($\text{m}^3 \text{ha}^{-1}$): net merchantable volume under bark by 10-year age class for even-aged stands.
- **Net Annual Increment** (NAI, $\text{m}^3 \text{ha}^{-1} \text{yr}^{-1}$): merchantable volume growth rate by age class, harmonised across countries using species-specific correction factors for bark, branches, top, and stump components.
- **Biomass Conversion and Expansion Factors** (BCEF, $t_{\text{DM}} \text{m}^{-3}$): age-class-resolved factors converting merchantable volume to total aboveground dry biomass, derived from species-specific allometric equations ([Boudewyn et al., 2007](#)).

The data are classified by country, administrative region (mostly NUTS-2), forest type (leading species), management type, and management strategy (even-aged or uneven-aged). The original NFI reference years span 1992–2018, with most countries reporting data from the 2005–2015 period.

Why this method. The yield-table-based approach was preferred over remote-sensing biomass products (e.g. [Avitabile et al., 2024](#)) because it: (i) provides region- and species-specific growth trajectories calibrated against national forest inventories; (ii) is consistent with the carbon accounting methodology used in the EU Carbon Budget Model (EU-CBM-HAT, [Grassi et al. 2018](#)); and (iii) yields a single rotation-averaged rate that is directly applicable as a long-run average CDR potential, which is arguably the parameter required

for modeling of the energy system. We applied the method only to productive forests assuming that new afforestation or reforestation project will fall within the categories: management types “H”, “HP”, “HS”, and “MAN”, while coppice, non-productive, and protected forests are excluded. The original method in [Fernandes et al. \(2026\)](#) derived the CO₂ sequestration rates from the standing above-ground biomass density (tDM / ha) of all forests, assuming a single Europe-wide averaged parameter as a single duration peak-growth-phase of forests.

0.1.2 Rotation-Averaged Sequestration Rate

For new afforestation, we compute a *rotation-averaged* Mean Annual Increment (MAI), which represents the long-term average CO₂ uptake rate over a full forest rotation. This approach avoids the overestimation that would result from using peak-growth-phase increment values, and is consistent with the EU LULUCF accounting framework ([European Parliament and Council, 2018, 2023](#)).

For each combination of forest type f , region r , and management type m , the computation proceeds as follows.

Step 1: Determine optimal rotation age. The rotation age T^* is defined as the age at which the volumetric MAI is maximised:

$$T^* = \arg \max_{T \geq T_{\min}} \frac{V(T)}{T} \quad (1)$$

where $V(T)$ is the standing stock ($\text{m}^3 \text{ha}^{-1}$) at age T and $T_{\min} = 25 \text{yr}$ is a minimum rotation age imposed to avoid numerical artefacts at very young age classes. The age classes are reported at 10-year intervals with midpoints at 5, 15, 25, $\dots$, 195 yr.

Step 2: Compute volumetric MAI. The rotation-averaged volumetric MAI is:

$$\text{MAI}_{\text{vol}} = \frac{V(T^*)}{T^*} \quad [\text{m}^3 \text{ha}^{-1} \text{yr}^{-1}] \quad (2)$$

Step 3: Convert volume to CO₂. The volumetric MAI is converted to a CO₂ sequestration rate using the BCEF at the rotation age, the IPCC carbon fraction, and a root-to-shoot expansion:

$$\text{MAI}_{\text{CO}_2} = \text{MAI}_{\text{vol}} \times \text{BCEF}(T^*) \times C_f \times \frac{M_{\text{CO}_2}}{M_C} \times (1 + R) \quad [\text{tCO}_2 \text{ha}^{-1} \text{yr}^{-1}] \quad (3)$$

where $\text{BCEF}(T^*)$ is the Biomass Conversion and Expansion Factor at the rotation age ($\text{t}_{\text{DM}} \text{m}^{-3}$), converting merchantable volume to total aboveground biomass; $C_f = 0.5$ is the carbon fraction of dry biomass ([IPCC, 2006](#)); $M_{\text{CO}_2}/M_C = 44/12$ is the molecular

mass ratio of CO₂ to carbon; and $R = 0.25$ is the root-to-shoot ratio accounting for belowground biomass (IPCC, 2006; Mokany et al., 2006).

Step 4: Regional aggregation. For each NUTS-2 region, the sequestration rate is computed as the equal-weight average across all productive forest types present in that region:

$$\overline{\text{MAI}}_{\text{CO}_2,r} = \frac{1}{|F_r|} \sum_{f \in F_r} \text{MAI}_{\text{CO}_2,f,r} \quad (4)$$

where F_r is the set of forest types with data in region r . Only even-aged, productive high-forest stands are included (management types “H”, “HP”, “HS”, and “MAN”).

0.1.3 Spatial Coverage: Mapping to All NUTS-2 Regions

The modeling approach introduced in Fernandes et al. (2026) requires sequestration rates for all the NUTS-2 region in the PyPSA-eur model which are later applied to the available land for afforestation computed from the CORINE LAND COVER dataset. The input library Pilli et al. (2024b) covers only 23 countries and uses a mix of NUTS-2, NUTS-1, and national-level regions. Gaps arise for countries outside the library (e.g. Norway, Switzerland, Turkey, the Western Balkans) as well as for specific NUTS-2 regions within covered countries (e.g. small island regions or city-states that fall below the spatial resolution of the NFI data).

To ensure full spatial coverage, a cascade of fallback strategies is applied in the following order:

1. **Direct NUTS-2 match:** the NUTS-2 region code in Pilli et al. (2024b) corresponds exactly to a NUTS-2 identifier.
2. **NUTS-1 propagation:** a three-character Pilli region code is matched to all child NUTS-2 regions sharing the same NUTS-1 prefix (e.g. DEA propagated to DEA1–DEA5).
3. **Country-level propagation:** for countries where the input library reports a single national value at NUTS-0 level, that value is applied uniformly to all NUTS-2 regions within the country.
4. **Distance-based neighbour mean:** for remaining unassigned regions, the mean of all NUTS-2 regions whose boundary lies within 100 km is used. A 100 km threshold was chosen to capture meaningful sea crossings—the Strait of Dover (≈ 34 km), the Irish Sea (≈ 70 km), and similar crossings in the Adriatic and Baltic—without introducing long-range links such as Scotland to Norway (≈ 300 km). The step is iterated until no further regions can be filled, propagating values continuously inward from continental anchors through island chains and isolated regions such as the British Isles.

5. **Country mean:** the mean of all resolved NUTS-2 regions within the same country, used for genuinely isolated island regions (e.g. the Canary Islands, Madeira, Azores).
6. **Mediterranean analogue (Malta, Cyprus):** MT00 and CY00 are assigned the value of EL43 (Crete), the geographically and climatically closest region with a direct Pilli estimate.

The source of each assigned value is recorded on zenodo (future) [Alamia \(2026\)](#) for reproducibility.

0.1.4 Temporal Weighting: Monthly Sequestration Profiles

To represent the seasonality of forest carbon uptake, the annual sequestration rate is disaggregated into monthly increments using observed Gross Primary Production (GPP) data as a proxy for the seasonal pattern of photosynthetic activity. GPP is the total amount of carbon fixed by plants through photosynthesis per unit area and time; it is highest in summer when light and temperature favour growth, and near zero in winter in temperate and boreal regions. It therefore serves as a direct observational indicator of when forests are actively sequestering carbon throughout the year.

Data source. Monthly weighting factors are derived from the FluxCom RS+METEO ensemble ([Jung et al., 2020](#)), which provides daily global GPP estimates at 0.5° spatial resolution using a machine-learning upscaling of eddy covariance flux tower observations, driven by ERA5 meteorological reanalysis and MODIS remote-sensing inputs. The specific dataset used is:

`GPP.RS_METEO.FP-ALL.MLM-ALL.METEO-ERA5.720_360.daily.{year}.nc`

Unit: $\text{gC m}^{-2} \text{ day}^{-1}$ Resolution: 0.5° global Years: 2010–2012 (≈ 3.6 GB total)

Source: [Jung et al. \(2020\)](#); data accessed via anonymous FTP at `ftp.bgc-jena.mpg.de` (Max Planck Institute for Biogeochemistry, Jena, Germany).

Three years (2010–2012) are used to construct a stable climatological seasonal cycle while limiting data volume. The ERA5-forced ensemble was selected over the purely remote-sensing-driven product because ERA5 provides continuous, gap-free meteorological forcing consistent with the climate data used elsewhere in PyPSA-Eur.

Why GPP as a temporal proxy. GPP is the appropriate proxy for the seasonal distribution of afforestation carbon uptake because: (i) it directly measures photosynthetic carbon assimilation, which drives biomass accumulation; (ii) it naturally captures the phenological seasonality of temperate and boreal forests, including growing-season onset and senescence; and (iii) it is available at the spatial and temporal resolution required for NUTS-2 aggregation across Europe.

Step 1: Monthly climatology. For each of the three downloaded annual files, the daily GPP field is clipped to the European geographical region, and the monthly mean climatological is computed by averaging all daily values within each month across all three years:

$$G_m(\lambda, \phi) = \frac{1}{N_m} \sum_{y=2010}^{2012} \sum_{d \in \mathcal{D}_{m,y}} \text{GPP}(\lambda, \phi, d) \quad [\text{gC m}^{-2} \text{ day}^{-1}] \quad (5)$$

where $\mathcal{D}_{m,y}$ is the set of days in month m of year y and N_m is the total number of averaging days. This yields a 12-layer climatological field at 0.5° resolution.

Step 2: Zonal aggregation to NUTS-2. Each 0.5° grid cell is assigned to a NUTS-2 region by a point-in-polygon test using the cell centroid and the NUTS-2013 polygon boundaries (Eurostat, 2013). For regions containing at least one grid cell, the monthly GPP is computed as the mean of all assigned cells with finite, positive values:

$$\bar{G}_{m,r} = \frac{1}{|\mathcal{C}_r^+|} \sum_{c \in \mathcal{C}_r^+} G_m(\lambda_c, \phi_c) \quad [\text{gC m}^{-2} \text{ day}^{-1}] \quad (6)$$

where $\mathcal{C}_r^+$ is the set of grid cells assigned to region r with $G_m > 0$. For very small regions (city-states, small islands) that contain no grid-cell centre, the value of the nearest grid cell (by centroid distance) is used as a fallback.

Step 3: Normalisation to monthly weights. For each NUTS-2 region, the 12 monthly GPP values are normalised to obtain dimensionless weights $w_{m,r}$ that sum to unity:

$$w_{m,r} = \frac{\bar{G}_{m,r}}{\sum_{m'=1}^{12} \bar{G}_{m',r}}, \quad \sum_{m=1}^{12} w_{m,r} = 1 \quad (7)$$

Months in which GPP is zero or not recorded (e.g. winter months in northern regions where the signal falls below the detection threshold) are treated as zero flux; if all 12 months are missing, the fallback cascade below is triggered instead of re-normalisation.

For regions where no valid GPP signal is available in any month, the same fallback cascade as in Section 0.1.3 is applied:

1. Mean profile of NUTS-2 regions within 100 km (EPSG:3035), iterated until stable, so that values propagate across sea crossings (e.g. southern England from northern France).
2. Mean profile of all resolved regions in the same country.
3. Uniform profile: $w_{m,r} = 1/12$ for all months (last resort).

Step 4: Monthly sequestration rates. The monthly CO₂ sequestration rate for region r is obtained by multiplying the annual rate by the monthly weight:

$$\text{Rate}_{m,r} = w_{m,r} \times \overline{\text{MAI}}_{\text{CO}_2,r} \quad [\text{tCO}_2 \text{ ha}^{-1} \text{ month}^{-1}] \quad (8)$$

By construction, $\sum_{m=1}^{12} \text{Rate}_{m,r} = \overline{\text{MAI}}_{\text{CO}_2,r}$, preserving the annual carbon budget. These monthly rates are passed to PyPSA-Eur as the time-varying efficiency of the afforestation CDR component.

0.1.5 Consistency with EU Policy Frameworks

The rotation-averaged MAI is consistent with the EU LULUCF Regulation ([European Parliament and Council, 2018, 2023](#)), which requires gross-net accounting of removals on afforested land over a 20–30 year transition period.

CRCF accounting. Regulation (EU) 2024/3012 ([European Parliament and Council, 2024](#)) established the EU Carbon Removal Certification Framework; the afforestation methodology was issued as a draft delegated act on 22 January 2026. Three points are relevant here.

(i) *Activity period:* 30 years (no renewal), followed by 40 years of monitoring — consistent with the PyPSA-Eur technology lifetime.

(ii) *Harvest is permitted* provided trees are replanted and no net stock reduction occurs (§1.1.3.2(h)(i)); a net decrease triggers a *reversal* and credit invalidation. The MAI concept is therefore appropriate: it already represents the long-run average annual stock increase of a managed stand that includes the harvest–replant cycle.

(iii) *Discounts.* Credits equal the net stock increase times $(1 - \text{UNC})$, where UNC is a mandatory uncertainty deduction (min. 10%). Operators also contribute a risk-rate share to a buffer pool after each audit; these units expire at end of monitoring (permanent discount). Risk rates from Table 7 of the draft are 8–17 % for broadleaf/mixed forests at low-to-medium disturbance. The net fraction of gross MAI that becomes usable credits is:

$$\eta_{\text{CRCF}} = (1 - \text{UNC})(1 - r_{\text{risk}}) \quad (9)$$

giving $\eta_{\text{CRCF}} \approx 0.75\text{--}0.83$ for typical European conditions. The remaining gross stock decomposes exactly as net certified stock (η_{CRCF}), buffer reserve $((1 - \text{UNC})r)$, and a harvest/safety margin (UNC) — carbon never credited and therefore extractable without reversal; the three sum to unity by construction. In PyPSA-Eur, η_{CRCF} is the **Link efficiency**, set via `afforestation[crcf_efficiency]` (default 0.80). Figure 1 illustrates all three components.

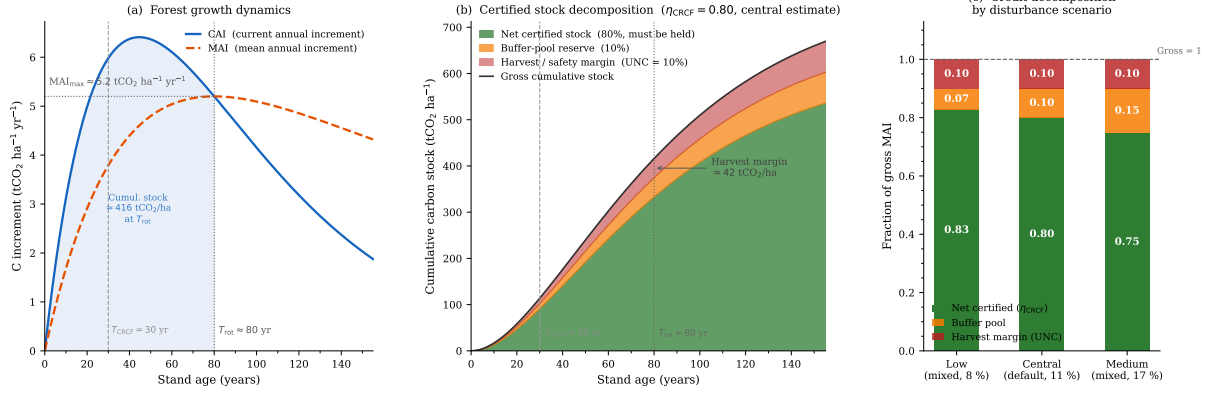


Figure 1: **Forest growth dynamics, certified stock, and CRCF credit breakdown.** (a) CAI (blue) and MAI (dashed orange) for a representative European managed forest ($T_{\text{rot}} \approx 80$ yr, $\text{MAI}_{\text{max}} \approx 5.2 \text{ tCO}_2 \text{ ha}^{-1} \text{ yr}^{-1}$; EU median from Pilli et al. 2024b); dashed line: 30-yr CRCF activity period. (b) Gross cumulative stock decomposed into net certified stock (green), buffer-pool reserve (orange), and harvest/safety margin (red, = UNC fraction, extractable without triggering reversal). Central estimate: $\eta_{\text{CRCF}} = 0.80$, $\text{UNC} = 10\%$, $r = 11\%$. (c) Same fractions for three disturbance scenarios (Table 7, draft delegated act, 22 Jan. 2026).

0.1.6 PyPSA-Eur implementation

Each network node has a `Store` on a dedicated `co2 afforestation` bus, linked to `co2 atmosphere` via an extendable `Link`. Storage capacity (`e_nom_max`) is set by the CORINE land area and annual sequestration rate; capital cost is annualised over a 30-year lifetime. The `Link` efficiency equals η_{CRCF} (Eq. 9), so only the certified fraction of gross sequestration reaches the storage bus. Both methods are available via `afforestation[potential_type]` ("density": Fernandes et al. 2026; "growth": this work; see https://github.com/BertoGBG/pypsa-eur/tree/a_CDRs).

Seasonal dispatch profile. In growth mode, $p_{\text{min_pu}}(t) = p_{\text{max_pu}}(t)$ forces the `Link` to follow the GPP-derived seasonal profile exactly, leaving the optimiser only the capacity p_{nom} to choose:

$$p_{\text{min},r}(t) = p_{\text{max},r}(t) = w_{m(t),r} \cdot \frac{T_{\text{year}}}{H_{m(t)}} \quad (10)$$

where $w_{m,r}$ is the normalised GPP weight for node r in month m (Eq. 7) and H_m is the number of snapshot hours in month m . The factor T_{year}/H_m preserves the annual budget: $\sum_t p_{\text{nom}} \bar{p}_r(t) = p_{\text{nom}} T_{\text{year}}$, equivalent to fixed monthly budgets proportional to $w_{m,r}$. The `Store` `e_nom_max` = Π_r ($\text{tCO}_2 \text{ yr}^{-1}$) implicitly caps the link: $p_{\text{nom}} \leq \Pi_r/T_{\text{year}}$. The optimiser therefore chooses the fraction of land to deploy; the seasonal shape is fixed by the profile.

0.1.7 Results and Validation

The computed rotation-averaged sequestration rates range from 2.1–12.6 tCO₂/ha/yr across 114 NUTS-2 regions with direct Pilli data, with a European median of 5.2 tCO₂/ha/yr and mean of 5.5 tCO₂/ha/yr. These values are consistent with the 5–15 tCO₂/ha/yr range reported for temperate European afforestation in the literature (Avitabile et al., 2024; Nabuurs et al., 2013).

As a validation case, the five Danish NUTS-2 regions yield rates of 3.9–5.8 tCO₂/ha/yr, with a national average of approximately 5.1 tCO₂/ha/yr. This is in reasonable agreement with the 6.5 tCO₂/ha/yr estimated by Fernandes et al. (2026) for a 60% conifer / 40% broadleaf species mix; the difference is attributable to the equal-weight species averaging used here versus an area-weighted approach giving greater weight to higher-productivity conifer species.

Table 1 reports a selection of regional rates.

Table 1: Rotation-averaged CO₂ sequestration rates for selected NUTS-2 regions. $\bar{T}$: mean rotation age across forest types included; n : number of productive forest types.

Country	NUTS-2	Rate [tCO ₂ ha ⁻¹ yr ⁻¹]	$\bar{T}$ [yr]	n
Austria	AT21	6.8	47.0	8.0
Belgium	BE20	6.9	34.0	7.0
Czechia	CZ01	5.2	48.0	4.0
Denmark	DK01	5.8	39.0	7.0
Denmark	DK05	3.9	39.0	7.0
Finland	FI1A	2.5	25.0	3.0
France	FRF1	5.4	27.0	6.0
Germany	DE2	7.2	34.0	8.0
Ireland	IE	7.1	33.0	6.0
Italy	ITH5	5.6	40.0	14.0
Poland	SL	5.9	31.0	8.0
Romania	RO21	6.0	28.0	9.0
Sweden	SE11	4.3	100.0	4.0
Sweden	SE33	2.1	33.0	4.0

Figure 2 shows the resulting monthly sequestration rates aggregated to the 50-node PyPSA-Eur network. The seasonal pattern is consistent with expectations: rates are near zero across most of Europe in winter (December–February), rise sharply in spring, peak in June–July for temperate and boreal nodes, and decline again in autumn. Southern nodes (Iberian Peninsula, Italy, Greece) exhibit a broader active season and a relatively flatter seasonal cycle, reflecting the year-round photosynthetic activity captured by the FluxCom GPP climatology in Mediterranean climates. The highest monthly rates (≈ 1.5 tCO₂ ha⁻¹ month⁻¹) occur in central European nodes during peak summer, consistent with the high rotation-averaged MAI values reported for Germany, Austria, and

Ireland in Table 1.

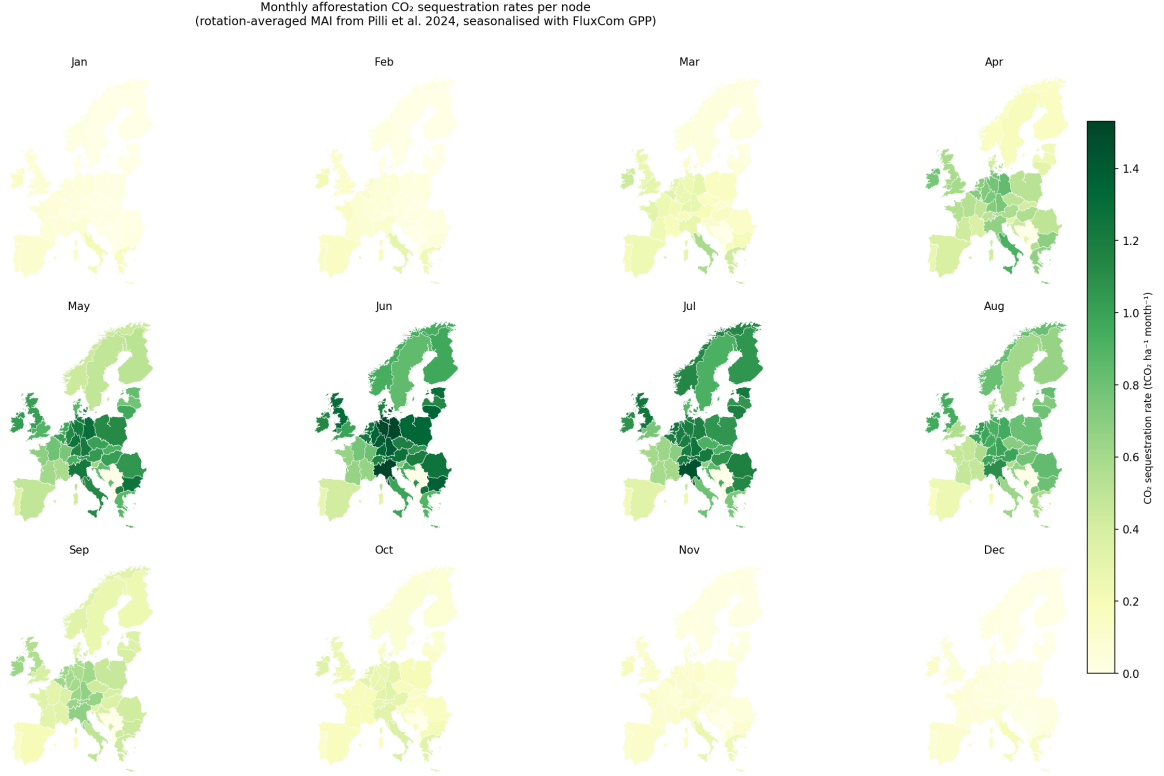


Figure 2: Monthly afforestation CO₂ sequestration rates per network node for a 50-node PyPSA-Eur configuration [tCO₂ ha⁻¹ month⁻¹]. Each panel shows one calendar month; node colours reflect the product of the rotation-averaged MAI (Section 0.1.2) and the GPP-derived monthly weight $w_{m,r}$ (Eq. 8). The strong north–south gradient in summer activity and the near-zero winter rates in boreal and continental nodes are clearly visible. Generated by `build_afforestation_potentials.py` in the PyPSA-Eur afforestation branch.

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